

*****Draft*****
Relook Ultibo
05/08/26
*****Draft*****

In 2013 I tried working with FPGA.

see

https://github.com/develone/zedboard_yocto/blob/master/doc/Zedboard_Yocto_test.pdf

see

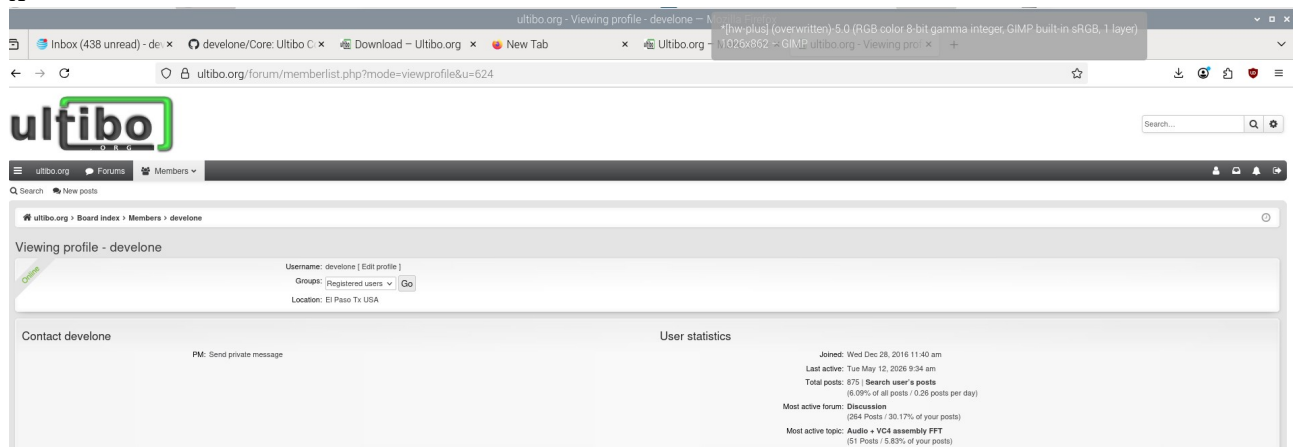
<https://github.com/develone/07118catzip/blob/master/doc/zipcpu.pdf>

see

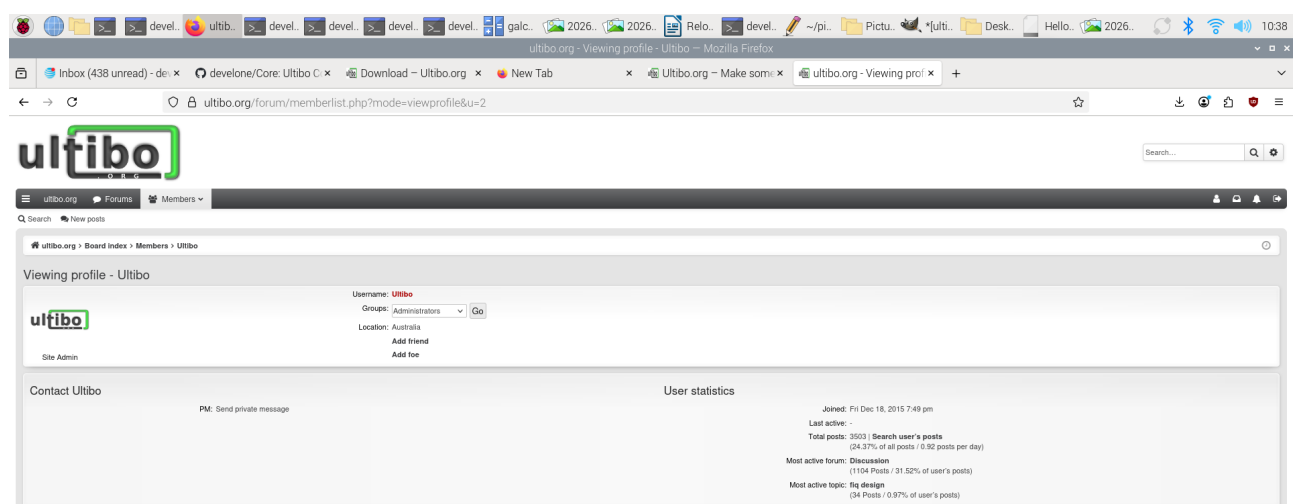
<https://github.com/develone/102121icozip/blob/catzip/doc/Working-Software-092821.pdf>

I started working with Ultibo since 2016.

X



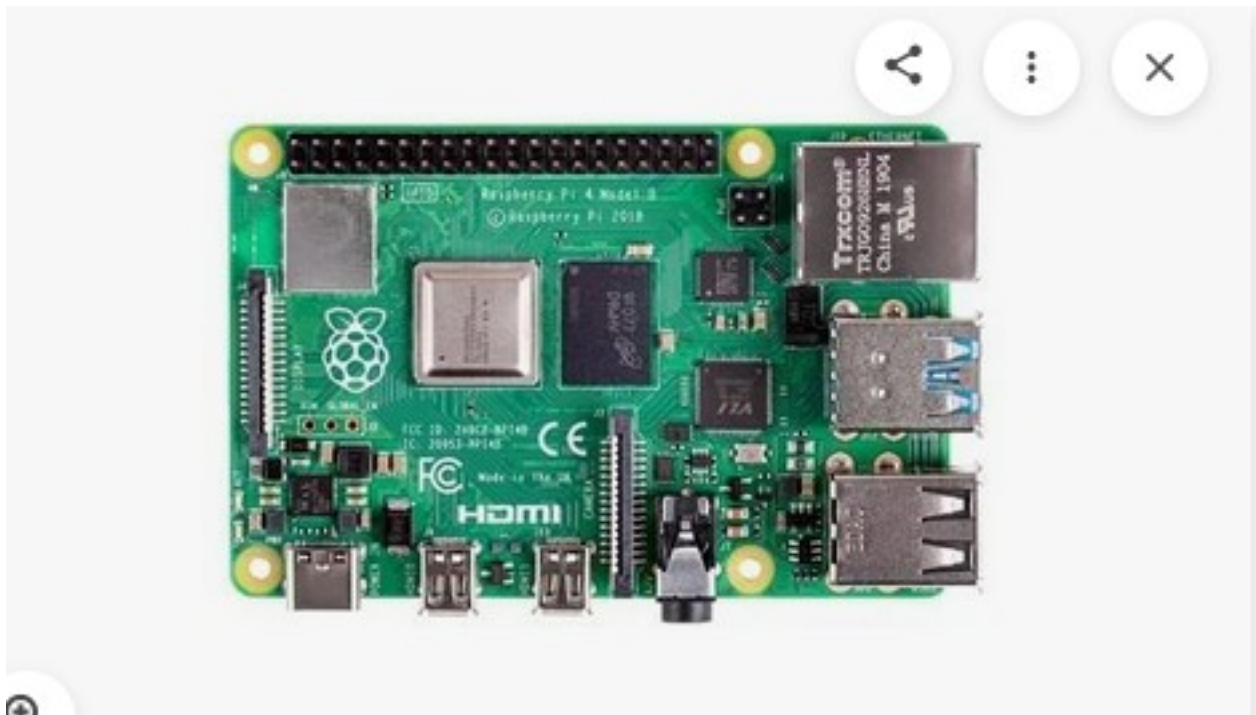
This was shortly after the projected started in 2015,



X

2025-11-24-raspbios-bookworm-arm64.img

Installed bookworm release on Raspberry Pi4 4Gb pi4-50



x

```
devel@pi4-50:~ $ shasum extra64.sh
1d92b567c492321c21ce7baaeebfd7e0cf83db3d  extra64.sh
```

```
curl https://pyenv.run | bash
```

05/12/26

```
devel@pi5-90:/media/devel/1b763776-4e1d-499c-9f24-a116a58c161f $ ls 64bit-pi4-26/
```

```
extra_pkgs.sh  extra_pkgs.sh.orig
```

```
installed-openocd082722-228ede-64bit.img
```

```
openocd082722-228ede-64bit.img
```

```
qemu-6.2.0-64bit.img  qemu-6.2.0-rpios-64bit.img
```

```
devel@pi4-50:~ $ sudo unsquashfs -d qemu-6.2.0 qemu-6.2.0-64bit.img
```

```
devel@pi4-50:~ $ sudo unsquashfs -d qemu-6.2.0-rpios qemu-6.2.0-rpios-64bit.img
```

```
qemu.sh
```

```
#!/bin/bash
```

```
export QEMU_PATH=$HOME/qemu-6.2.0-rpios/bin
```

```
export PATH=$QEMU_PATH:$PATH
```

```
echo $PATH
```

```
devel@pi4-50:~ $ ./ultiboinstaller.sh
```

Linux installer for Free Pascal and Lazarus (Ultibo edition)

This installation will download the sources for:

Ultibo core

Ultibo tools

Ultibo examples

Free Pascal (Ultibo edition)

Lazarus IDE (Ultibo edition)

Then it will build all of the above, this will take several minutes to complete depending on the speed of your system.

The installation will not interfere with any existing development environments including other installations of Free Pascal and Lazarus.

Continue (y/n)?

Do you want to build and install the Lazarus IDE (y/n)?

These can be installed on Debian based distributions using:

```
sudo apt-get install build-essential gdb-minimal unzip
```

Lazarus IDE requires the GTK2 and X11 dev packages which can be installed on Debian based distributions by using:

```
sudo apt-get install libgtk2.0-dev libcairo2-dev \
  libpango1.0-dev libgdk-pixbuf2.0-dev libatk1.0-dev \
  libghc-x11-dev
```

(On some distributions libgdk-pixbuf2.0-dev might have been replaced by libgdk-pixbuf-xlib-2.0-dev)

Cross compiling Ultibo applications from Linux requires the arm-none-eabi and aarch64-linux-gnu builds of the binutils package, these can be installed on Debian based distributions using:

```
sudo apt-get install binutils-arm-none-eabi \
  binutils-aarch64-linux-gnu
```

Press return to check for these prerequisites

Enter an installation folder or press return to accept the default install location

[/home/devel/ultibo/core]:

The install folder will be:

/home/devel/ultibo/core

Continue? (y,n):

After install do you want a shortcut created in:

/home/devel/.local/share/applications (y/n)?

Do you want to build the Hello World examples (y/n)?

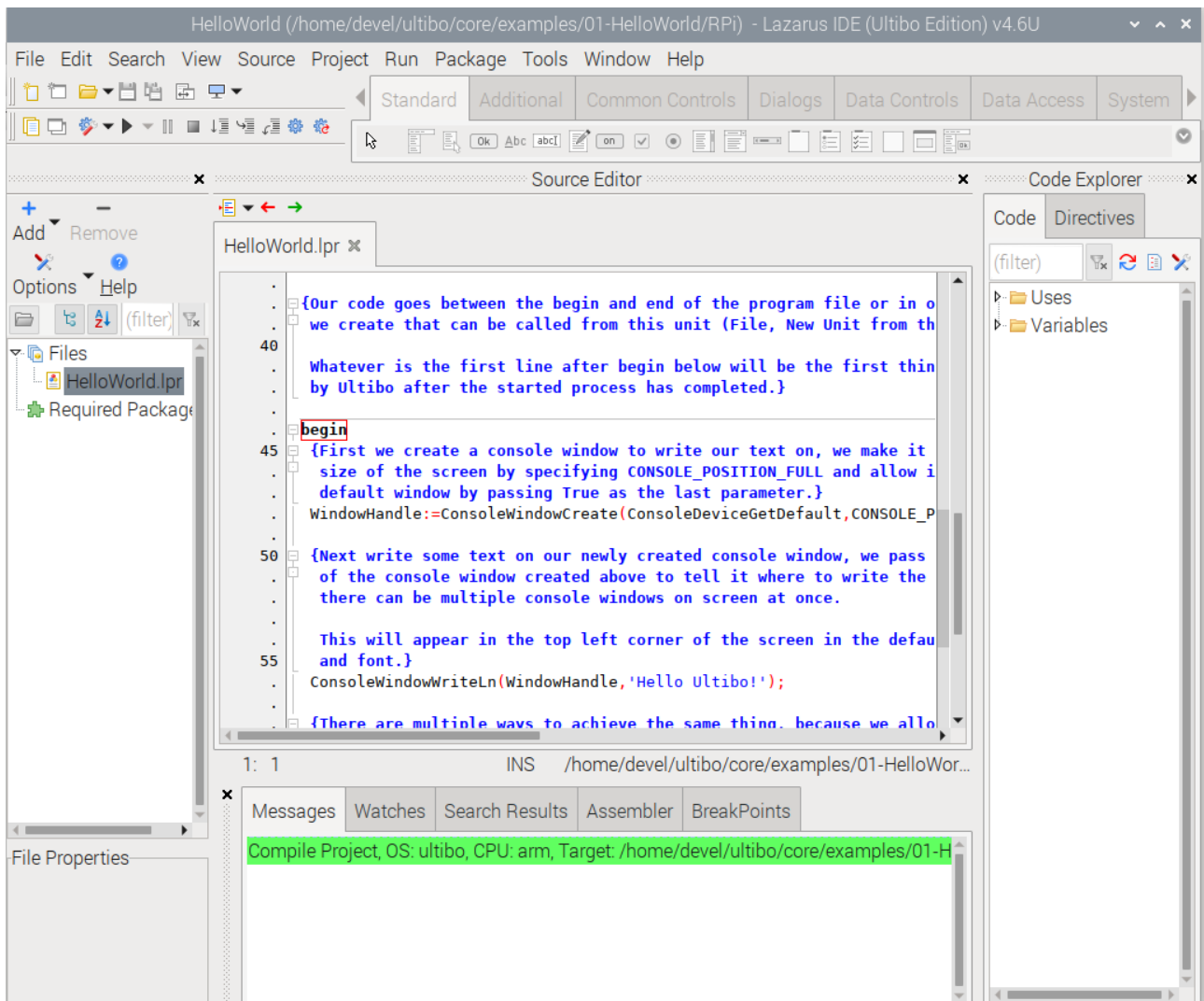
Tested Raspbery Pi-Zero

x



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This is an example of the HelloWorld for the Raspberry Pi-Zero.



Green bar indicates that the program compiled ok.

X

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devel@pi4-50:~/Ultibo_Projects/HelloWorld-plus/RPi \$ diff HelloWorld.lpr

~/ultibo/core/examples/01-HelloWorld/RPi/HelloWorld.lpr

57,62c57

< ConsoleWindowWriteLn(WindowHandle,'This program was compiled using');

< ConsoleWindowWriteLn(WindowHandle,'on a Raspberry Pi4 4GB');

< ConsoleWindowWriteLn(WindowHandle,'running Lazarus IDE Ultibo Edition');

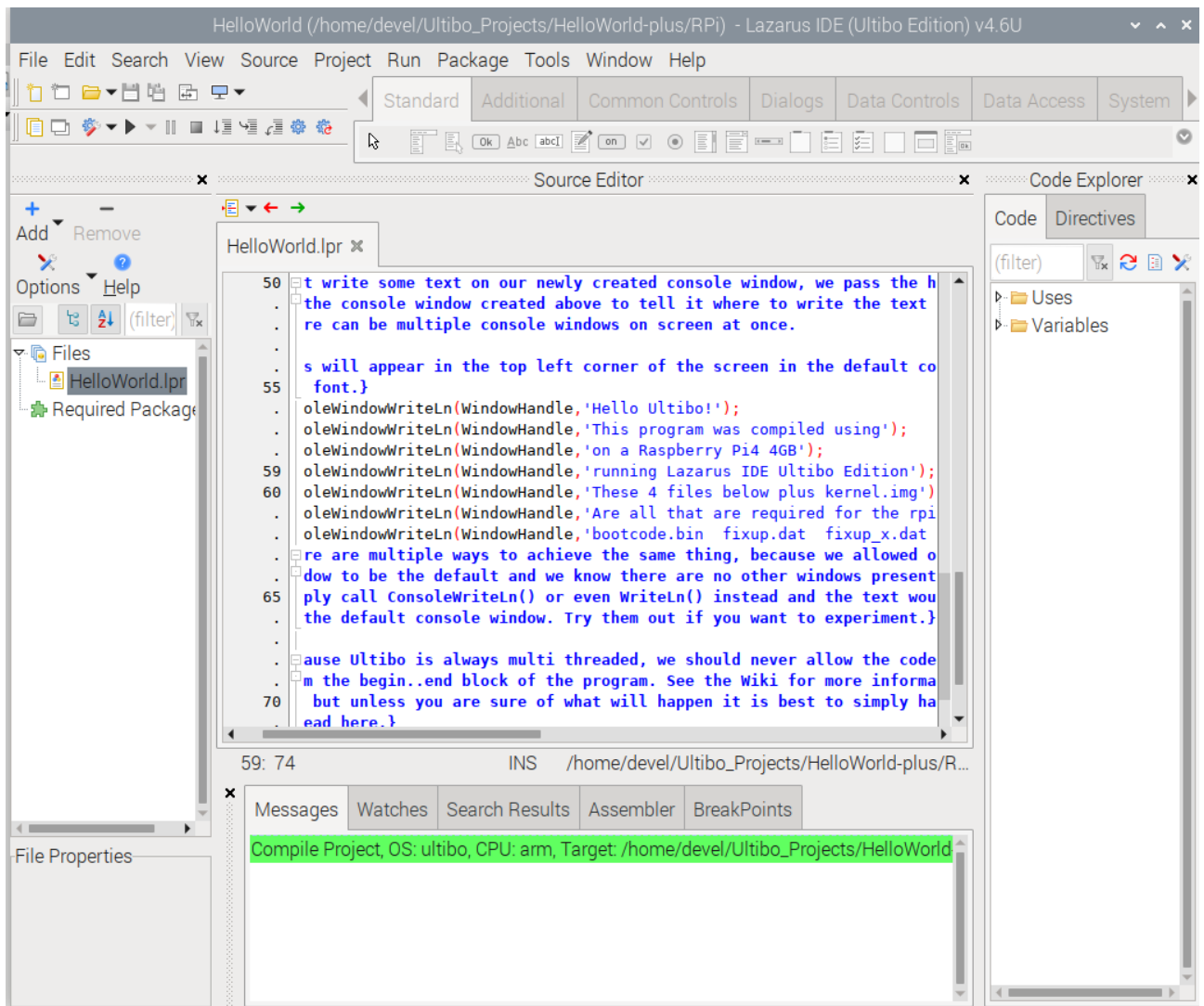
< ConsoleWindowWriteLn(WindowHandle,'These 4 files below plus kernel.img');

< ConsoleWindowWriteLn(WindowHandle,'Are all that are required for the rpi-zero');

< ConsoleWindowWriteLn(WindowHandle,'bootcode.bin fixup.dat fixup_x.dat start4x.elf
start.elf');

>

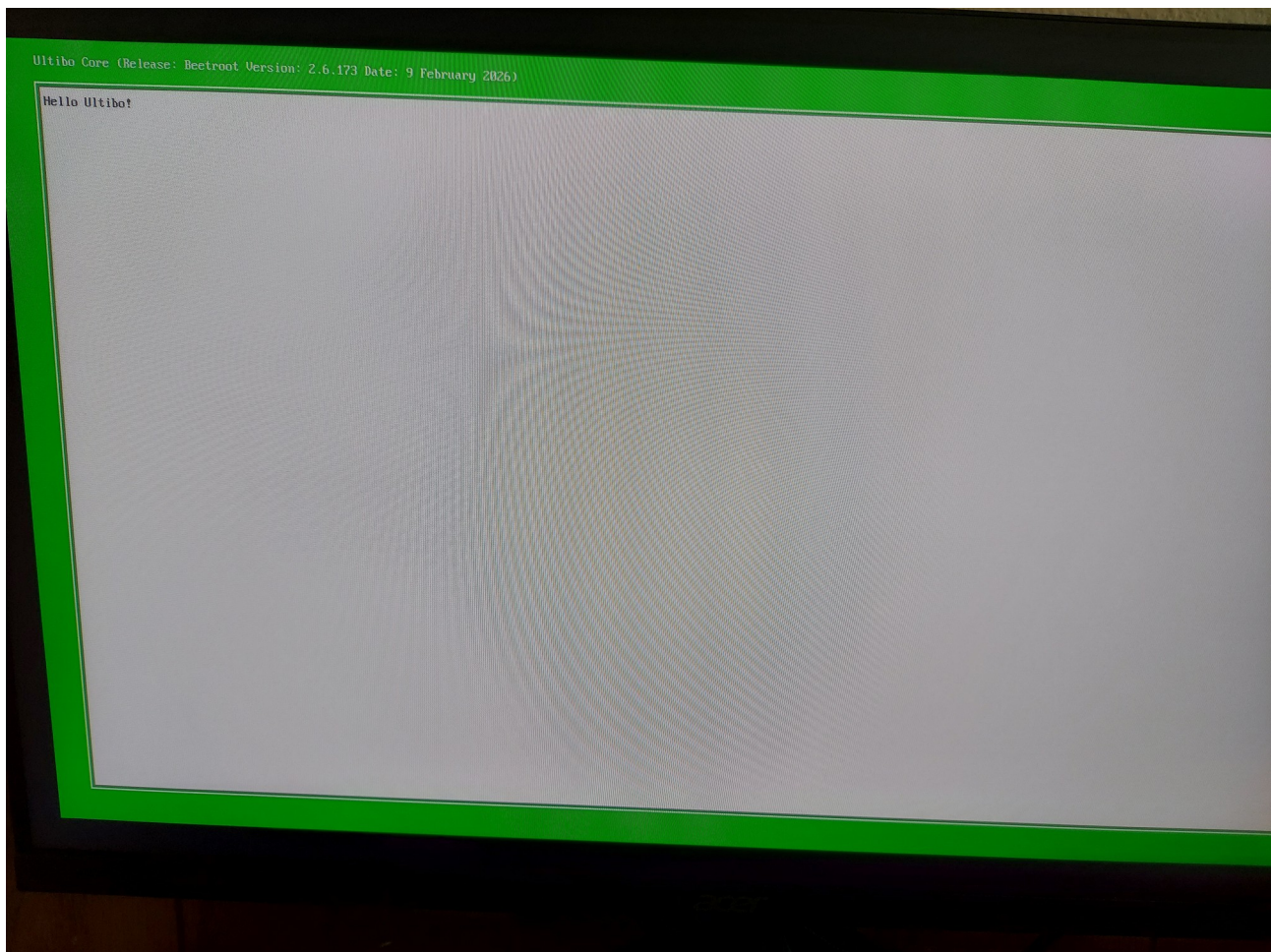
Green bar indicates that the program compiled ok.



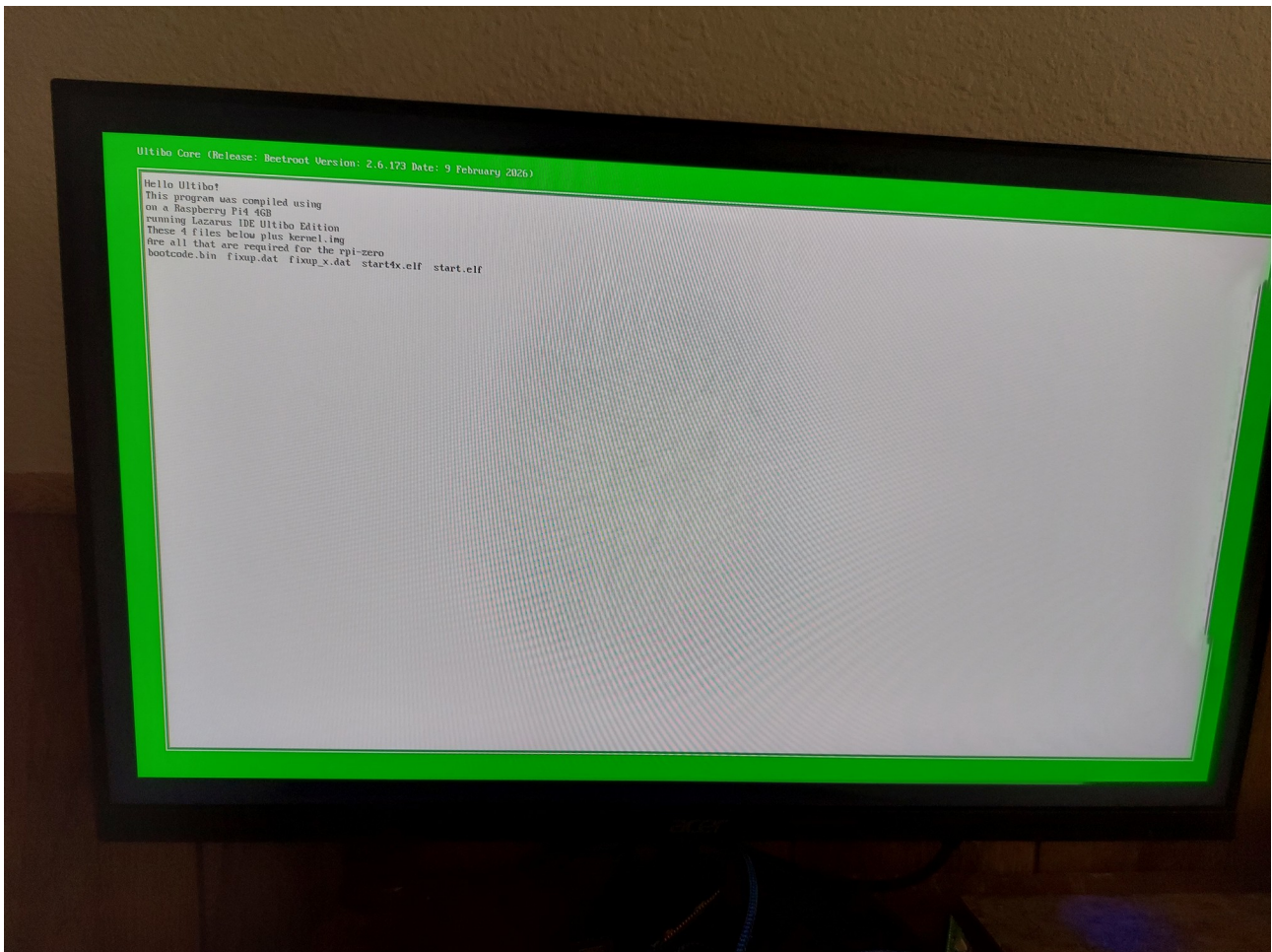
```

devel@pi4-50:~/ultibo/core/examples/01-HelloWorld/RPi $ ls ~/firmwar_for_ultibo/112221/
bootcode.bin fixup.dat fixup_x.dat start4x.elf start.elf plus kernel.img

```

```
devel@pi4-50:~/ultibo/core/examples/01-HelloWorld/RPi $ ls ~/firmwar_for_ultibo/112221/  
bootcode.bin  fixup.dat  fixup_x.dat  start4x.elf  start.elf  plus  kernel.img
```



X



Raspberry Pi 2 Model B Desktop (Quad Core CPU 900 MHz, 1 GB RAM, Linux)

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CPU Model	Cortex A7

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```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/src $ ./compile_ultibo.sh
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/src $ ls *.c | wc
  22   22  182
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/src $ ls *.c
bio.c      invert.c  opj_clock.c  t2.c
cio.c      j2k.c     opj_malloc.c  tcd.c
dwt.c      jp2.c     pi.c          tgt.c
event.c    mct.c     rd-wr-ops.c  thread.c
```



```
function_list.c mqc.c    sparse_array.c
image.c        openjpeg.c t1.c
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU $ ./libbuild.sh
```

```
dwtlift.c: In function 'decompress':
```

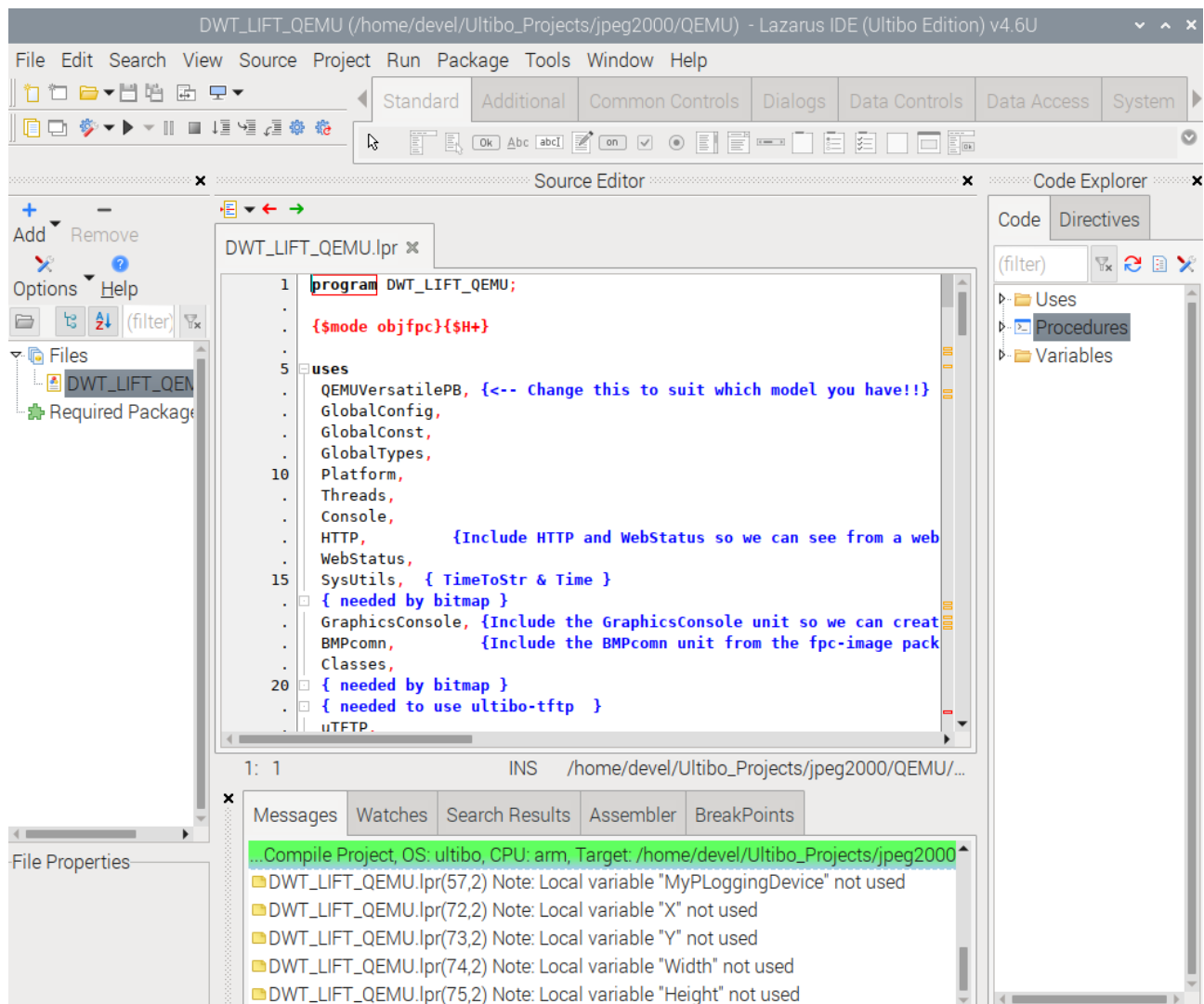
```
dwtlift.c:658:17: warning: implicit declaration of function 'octave_write_byte' [-Wimplicit-function-declaration]
```

```
658 |         octave_write_byte(r_decompress_fn,r_decompress,da_x1*da_y1);
    |         ^~~~~~
```

when ./libbuild.sh is executed should be 23

```
23 23 192 libdwtlift_obj.txt
```

X



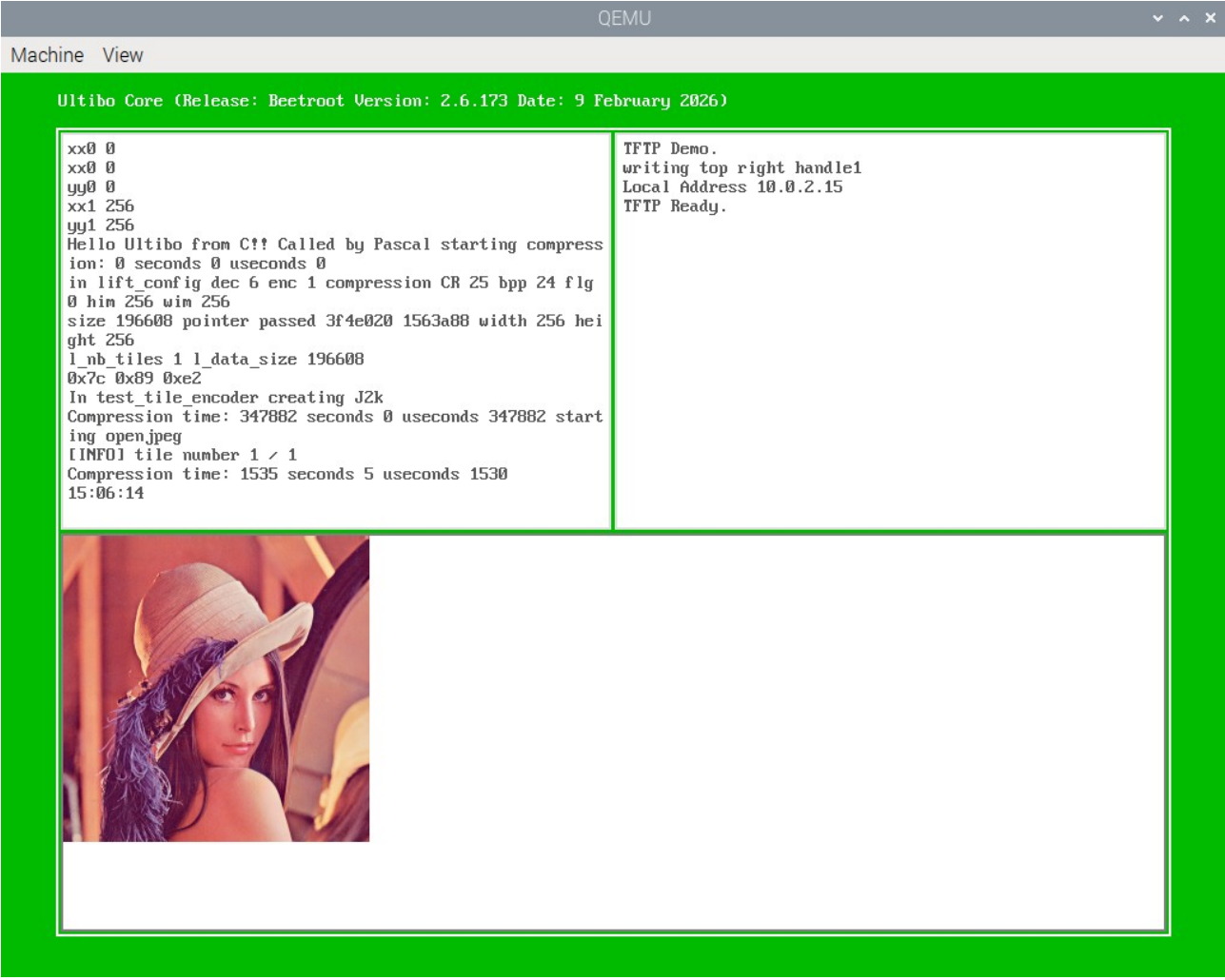
X

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU $ . ~/qemu.sh
```

```
/home/devel/qemu-6.2.0-rpios/bin:/home/devel/.pyenv/plugins/pyenv-virtualenv/shims:/home/
devel/.pyenv/shims:/home/devel/.pyenv/bin:/home/devel/local/openocd/bin:/usr/local/sbin:/usr/
local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/local/games:/usr/games
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU $ ./startqemu.sh
```

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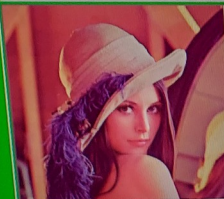
X

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bo Core (Release: Beetroot Version: 2.6.173 Date: 9 February 2026)

```
xyz
p 24
ze 196608
inter passed f20020
. 65536.000000 sqrt tt 256.000000 256 256
scal char ptr f20020
llocating memory with malloc
mg->n_red 0xc00004f0
mg->n_green 0xc00004f0
mg->n_blue 0xc00004f0
mg->n_tnp 0xc00004f0
copying RGB 8 bit char to 32 int
mg->n_red 0xc00004f0
mg->n_green 0xc00004f0
mg->n_blue 0xc00004f0
reseting pointers
mg->n_red 0xc00004f0 passed ptr 0xf20020
mg->n_green 0xc00004f0
mg->n_blue 0xc00004f0
Calling lifting red
Calling lifting green
Calling lifting blue
lifting to Buffer
Elapsed time: 34412 microseconds
156
01:39:29
```

```
writing top right handle1
Buffer 00f20020 Size 196608 LineSize 768 BitCount 24
UPSIDEDOWN
Going to free Buffer memory
Local Address 192.168.12.161
TFTP Ready.
```



Bitmap file saved successfully

